ζ_{df} 0,2

1. $n=10$ winters #Avalanches $= [2,0,1,3,1,0,2,1,4,1]^T$

a) $P(\Lambda=\lambda) \sim \text{Poisson}(\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$

$\Rightarrow L(\lambda) = \prod_{i=1}^{n} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{x_1+x_2+\dots+x_n} e^{-n\lambda}}{x_1! \cdot x_2! \cdots x_n!}$

$L(\lambda) = \frac{(10\lambda)^{15} e^{-10\lambda}}{15!}$

$\text{Gamma}(a,b) = \sum_{i=1}^{a} b e^{bt} = \frac{b^a x^{a-1} e^{-bx}}{\Gamma(a)}$

$P(H|D) = \frac{P(D|H) \times P(H)}{P(D)}$ ($P(D)$: Poisson(λ); $P(H)$: Gamma(a,b))

$\text{Gamma}(a,b)\, L(\lambda) = \frac{b^a x^{a-1} e^{-bx}}{\Gamma(a)} \cdot \prod_{i=1}^{n} \frac{\lambda_i^{x_i} e^{-\lambda}}{x_i!}$ $\quad \lambda = b$

$= \frac{b^a \lambda^{\sum x_i} e^{-n\lambda} e^{-bx} x^{a-1}}{\Gamma(a) \sum x_i}$

$= \frac{b^a b^x e^{-(n+b)} x^{a-1}}{\Gamma(a+x)} = \frac{b^{a+x} e^{-n+b}}{\Gamma(a+x)} \propto \text{Gamma}(a+x, b+n)$

1 b) $$P(Y=k)=\int_0^\infty \frac{e^{-\lambda t}(\lambda t)^k}{k!}\cdot\frac{b'^{a'}\lambda^{a'-1}e^{-b'\lambda}}{\Gamma(a')}\,d\lambda$$

take out constants $$\frac{b'^{a'}t^k}{k!\Gamma(a')}\int_0^\infty e^{-\lambda(b'+t)}(\lambda)^{a'-1+k}\,d\lambda$$

Integration by parts $$\frac{b'^{a'}t^k}{k!\Gamma(a')}\left[\frac{-1}{b'+t}\lambda^{a'-1+k}e^{-(\lambda)(b'+t)}+\frac{a'-1+k}{b'+t}\int\lambda^{a'-2+k}e^{-\lambda(b'+t)}\,d\lambda\right]_0^\infty$$

again $$=\frac{b'^{a'}t^k}{k!\Gamma(a')}\left(\frac{a'-1+k}{b'+t}\right)\left(\frac{-1}{b'+t}\lambda^{a'-2+k}e^{-\lambda(b'+t)}+\frac{a'-2+k}{b'+t}\int\lambda^{a'-3+k}e^{-\lambda(b'+t)}\,d\lambda\right)_0^\infty$$

$$=\frac{b'^{a'}t^k}{k!\Gamma(a')}\cdot\frac{a'(a'-1+k)(a'-2+k)}{(b'+t)^3}\int\lambda^{a'-3+k}e^{-\lambda(b'+t)}\,d\lambda$$

Continuing reduction yields $\Rightarrow$ (for k iterations)

$$\frac{\Gamma(a'+k)}{\Gamma(a')k!}\left(\frac{b'}{b'+t}\right)^{a'}\left(\frac{t}{b'+t}\right)^k \sim \text{Binomial}\left(a'+k,\frac{b'}{b'+t}\right)$$

$n=a'+k$

Binomial distribution

$E[\text{Binomial}]=n\cdot p=(a'+k)\left(\frac{b'}{b'+t}\right)$

$\text{Var}(\text{Binomial})=n\cdot p(\bar{p})=(a'+k)\left(\frac{b'}{b'+t}\right)\left(\frac{t}{b'+t}\right)$

$E[\text{Poisson}(\lambda t)]=\lambda t$

$\text{Var}(\text{Poisson}(\lambda t))=\lambda t$

the Binomial does not rely on rate parameter λ

1 b) for $t=1$

$$P(Y=0) = \text{Poisson}(2\cdot1) = \frac{2^0 e^{-2}}{0!} = e^{-2} = .1353$$

$$\text{Binomial} = \frac{\Gamma(19+0)}{\Gamma(19)\,0!}\left(\frac{12}{12+1}\right)^{19}\cdot\left(\frac{1}{12+1}\right)^{0} = .2135$$

$t=5$

$$P(Y=0) = \text{Poisson}(2\cdot5) = \frac{2^5 e^{-2}}{5!} = .0017467$$

$$\text{Binomial} = \frac{\Gamma(19+0)}{\Gamma(19)\,0!}\left(\frac{12}{12+5}\right)^{19}\cdot\left(\frac{5}{12+5}\right)^{0} = .001336$$

accuracy increases as t increases

$$P(Y\geq10) = 1-P(Y<10) = 1-\sum_{i=0}^{9}\frac{\lambda^i e^{-\lambda}}{i!} = 1-\sum_{i=0}^{9}\frac{(10)^i e^{-10}}{i!} = .5421$$

$$\text{Binomial} = 1-\sum_{i=0}^{9}\frac{\Gamma(19+i)}{\Gamma(19)\,i!}\left(\frac{12}{17}\right)^{19}\cdot\left(\frac{5}{17}\right)^{i} = .0619$$

Poisson

$$EM = \text{Var} = 5\cdot2 = 10$$

Binomial

$$EM = n\cdot p = (19+10)\left(\frac{12}{17}\right) = 20.47 \qquad (19+10)\left(\frac{5}{17}\right) = 8.529$$

$$\text{Var} = n\cdot p\cdot q = 20.47\left(\frac{5}{17}\right) = 6.02076$$

Differences likely come from not enough trials since Binomial only is approximated by Poisson for large n

1. (i) Since $H=20$ _ Binomial $\sim$ Poisson(λH)

theorem 2.5 If n is large, binomial$(n, \lambda/n)$ distribution is approximately poisson(λ)

$$\frac{\lambda^k}{k!} \frac{n(n-1)\cdots(n-k+1)}{n^k} \left(1-\frac{\lambda}{n}\right)^n \left(1-\frac{\lambda}{n}\right)^{-k}$$

i) $\frac{\lambda^k}{k!}$ not dependent on n

ii) k terms on top and bottom

$\Rightarrow \frac{n}{n} \frac{n-1}{n} \cdots \frac{n-k+1}{n}$

iii) $\left(1-\lambda/n\right)^n \to e^{-\lambda}$ $n \to \infty$

$\Rightarrow \frac{\lambda^k}{k!} \cdot 1 \cdot e^{-\lambda}$ Poisson(λ)

$\Rightarrow$ Binomial$(20, .02 = \lambda/n) \Rightarrow \lambda = .02(20) = .4$

$\Rightarrow P(\text{destruction}=0 \mid \lambda) = \frac{(\lambda H)^0}{0!} e^{-\lambda H} = e^{-\lambda H} = e^{-.4(20)} \approx .000335$

$$\int_6^\infty \lambda^{a'-1} e^{-(b'+s)\lambda} \, d\lambda = \Gamma(a')/(b'+s)^{a'} = \frac{14!}{(12+20)^{14}} = .00033303$$

$.00033303 \approx .000335$

Yes, deflection wall should be built.

3. $\lambda(t) = 12 + 8\cos\left(\frac{\pi(t-16)}{12}\right)$

2PM = 14hours 6PM = 18 hours
3am = 3hours 4am = 4hours

a) N = # of people $\sim$ Poisson(λt)

$E[t] = \Lambda(t) = \frac{dE[t]}{dt}$

$\Rightarrow dE = \lambda(t)\,dt$

$E[\#] = \int_{t_1}^{t_2} \lambda(t)\,dt = \int_{t_1}^{t_2} 12 + 8\cos\left(\frac{\pi(t-16)}{12}\right) dt = \left[12t + \frac{96}{\pi}\sin\left(\frac{\pi(t-16)}{12}\right)\right]_{t_1}^{t_2}$

1 day

$E[t] = 12\cdot 24 + \frac{96}{\pi}\sin\left(\frac{2\pi}{3}\right) - \frac{96}{\pi}\sin\left(-\frac{4\pi}{3}\right)$

$\sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$, $\sin\left(-\frac{4\pi}{3}\right) = \frac{-\sqrt{3}}{2}$

$E[\#] = 288 + 43.3117 = 331.3117$

2PM – 6PM

$E[t] = 12\cdot 18 + \frac{96}{\pi}\sin\left(\frac{\pi}{6}\cdot\right) - \left(12\cdot 14 + \frac{96}{\pi}\sin\left(-\frac{\pi}{6}\right)\right)$

$E[t] = 73.5577$

3am–4am

$E[t] = 12\cdot 4 + \frac{96}{\pi}\sin\left(-\frac{\pi}{3}\right) - \left(12\cdot 3 + \frac{96}{\pi}\sin\left(\frac{-\pi}{4}\right)\right)$

$E[t] = 7.1435$

$E[t] = \int \lambda\,dt = \Lambda(3,4)$
$\Rightarrow E[t] = 7.1438$

$P(N>10) = 1 - P(N\le 10) = 1 - \sum_{i=0}^{10} \frac{(E[t])^i e^{-E[t]}}{i!}$

$= 1 - \sum_{i=0}^{10} \frac{(7.14)^i e^{-7.1435}}{i!} = 1 - .891 = .1090 = P(N>10)$

If manager uses $\lambda = 12$

they would predict between 2 and 6 PM 24 arrivals, way off from 78 expected arrivals

and 12 arrivals between 3 and 4 am, also way off from expected 1.14 arrivals

they would predict $P(N \geq 10) = 1 - \sum_{i=0}^{10} \frac{(12)^i e^{-12}}{i!}$

$P(N > 10) = .6528$, Way different than the previous .10 calculated

b) $\lambda_{max} = 20$ $\quad N \sim \text{Poisson}(20(t-s))$

$$P(T) = \frac{\lambda(T)}{20}$$

$$P(s < T \le t) = \int_s^t \frac{\lambda(T)}{20} dT = \frac{\Lambda(s,t)}{20} \cdot \frac{1}{t-s} = \frac{\Lambda(s,t)}{20(t-s)}$$

$$= \frac{12(t-s) + \frac{96}{\pi}\left(\sin\left(\frac{\pi(t-16)}{12}\right) - \sin\left(\frac{\pi(s-16)}{12}\right)\right)}{20(t-s)}$$

$$P = P_{gen} \cdot P_{kept} = P_{Poisson}(20(t-s)) \, B\text{inom}\left(\frac{\Lambda(t,s)}{20(t-s)}\right) = \text{Poiss}(\Lambda(t,s))$$

~~$P(N=k) = p^k q^{n-k}$~~ Theorem 2.5 If n is large enough binomial $(n, \lambda/n)$ distribution $\cong$ Poisson (λ)

fraction kept $= \frac{\int_0^t \lambda(t)/20}{}$

for a day $= \frac{\int_0^t \lambda(t) dt}{20} = \frac{\Lambda(0,24)}{20} \cdot \frac{1}{24} = \frac{381}{20} \cdot \frac{1}{24} = \frac{19}{24} = .79375$

3 c)

Simulation

$\bar{E} = 4.1$

almost identical

Var_Poisson = 2.026

significantly far off

since $E[\text{Poisson}(1)] = 1$

$\text{Var}[\text{Poisson}(1)] = 1$

E of $\text{Poisson}(4.041) = 4.041$

$\text{Var}(\text{Poisson}) = 4.041$

2. a) $N_p = N_s + N_b$

N_s = # Photons from star
N_b = # Photons from background
N_p = # Photons
$N_s \sim \text{Poisson}(3)$
$N_b \sim \text{Poisson}(2)$

$q = .4$

$$P(N_s + N_b = n) = \sum_{m=0}^{n} P(N_b = m) P(N_s = n-m)$$

$$= \sum_{m=0}^{n} e^{-2}\frac{2^m}{m!} e^{-3}\frac{3^{n-m}}{(n-m)!}$$

$$= e^{-3-2}\frac{(3+2)^n}{n!}\sum_{m=0}^{n}\binom{n}{m}\left(\frac{3}{3+2}\right)^{n-m}\left(\frac{2}{3+2}\right)^{m}$$

Since the sum is 1, and the sum of all probabilities is binomial on binomial(n, q), N_s and N_b are ind. Poisson processes

$$ET = \frac{1}{\lambda} \qquad ET_p = \frac{1}{3+2}\cdot\frac{1}{q} = \frac{1}{5}\cdot\frac{1}{.4} = .5 = ET$$

$$P(N_p = 0) = P(N_s = 0)P(N_b = 0) = \frac{e^{-3\cdot1.5}3^0}{0!}\cdot\frac{e^{-2\cdot1.5}2^0}{0!} = e^{-(5)(1.5)} = .01012$$

$$P(N_p \geq 0) = .01012$$

$$P(T < 1.5) = \int_0^{1.5} \lambda_{tot} e^{-\lambda_{tot} t}\, dt = \int_0^{1.5} 5e^{-5t}\, dt = \left[-e^{-5t}\right]_0^{1.5} = 1 - e^{-5} = .9933$$

$$P(T < 1.5) = .9933$$

~~$P(N_p = 3) = \frac{e^{-\lambda_{tot}\cdot 1.5}(\lambda_{tot}\cdot 1.5)^3}{3!} = \frac{e^{-5\cdot 1.5}(5\cdot 1.5)^3}{3!}(.6) = .3387$~~

~~$P(t \quad) =$~~

$$P(N_p \geq 3) = 1 - P(N_p \leq 3) = 1 - P(N_p = 2) - P(N_p = 1) - P(N_p = 0)$$

$$= 1 - \left(\sum_{i=0}^{2}\frac{e^{-\lambda_{tot} t}(\lambda_{tot} t)^{x_i}}{x_i!}\right) = 1 - .0095 = .441 = P(N_p \geq 3)$$

S = time Photon to hit from star b = time Photon to hit from background

b) $P_{min}(\min(S,b)>t) = P(S>t, b>t) = P(S>t)P(T>t)$

$= e^{-\lambda_S t} e^{-\lambda_b t} = e^{-(\lambda_S+\lambda_b)t}$

$P(S<b) = \int_0^\infty F_S(s) P(b>t)\,dt = \int_0^\infty \lambda_S e^{-(\lambda_S+\lambda_b)t}\,dt$

$= \frac{\lambda_S}{\lambda_S+\lambda_b}\int_0^\infty (\lambda_S+\lambda_b) e^{-(\lambda_S+\lambda_b)t}\,dt = \frac{\lambda_S}{\lambda_S+\lambda_b} = \frac{3}{3+2} = \frac{3}{5} = .6$

$P(N_S=6 \mid N_p=10) = P(N_S=6)\,P(N_b=4) = \frac{e^{-3.5}(3.5)^6}{6!}\,\frac{e^{-2.5}(2.5)^4}{4!}$

$P(N_S=6 \mid N_p=10) = .000041545$ if $N_p=10$, $N_b = N_p - N_S = 10-6 = 4$

arrival time and first detection are independent

$\Rightarrow P(S\le t, S<1) = P(S\le b)\,P(S<1) = \frac{\lambda_S}{\lambda_S+\lambda_b}\cdot\int_0^1 \lambda_S e^{-\lambda_S t}\,dt$

$= .6\left[-e^{-3t}\right]_0^1 = .6(1-e^{-3}) = .57013 = P(S>t, S<1)$

$\lambda_{tot} = 2+3 = 5 \Rightarrow E_p T = \frac{1}{5} = .2$

$P(S<b) = \frac{\lambda_S}{\lambda_S+\lambda_b} = .6 \quad P(b<S) = \frac{\lambda_b}{\lambda_S+\lambda_b} = \frac{2}{5} = .4$

$\Rightarrow E_p T + P(S<b)E_S T + P(b<S)E_b T = .2 + .6(.333) + .4(.5)$

$E_p T = .65$

c) Mean

simulation: $\mu_5 = 6.4127$, $\mu_6 = 4.4034$

Direct: $\mu_5 = 7.0033$, $\mu_6 = 5.0034$

Standard deviation

via simulation: $\sigma_5 = 2.4942$, $\sigma_6 = 2.0124$

Direct: $\sigma_5 = 2.4521$, $\sigma_6 = 2.0079$

Correlation

simulation:
$$\begin{bmatrix} 1.2117 & .0003 \\ .0003 & .8253 \end{bmatrix}$$

direct:
$$\begin{bmatrix} 1.242 & .0034 \\ .0034 & .8188 \end{bmatrix}$$

$E[\text{Poisson}(10)] = \lambda = 10$

$\mu_{5+6} = \mu_5 + \mu_6 = 4.4 + 6.4 = 10.8213$

$P(N_5 = 6 \mid N_+ = 10) = \frac{\#\, N_5 = 6}{\#\, N_p = 10} = \frac{61223}{24556} = .2534$

$P(5 \le 4) = \frac{\#\, 5 \le 4}{\text{tot}\,\#} \approx \frac{140000}{200000} = .4507$

$P(5 \le 6) = \frac{\#\, 5 \le 6}{\text{tot}\,\#} \approx \frac{120000}{200000} \approx .6024$

$E[M] = .8356$